

Venkatkrishnan Iyer

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Available for 2027 full-time roles & future internships

EDUCATION

University of Michigan

Bachelor of Engineering in Computer Engineering, Minor in Asian Languages

Ann Arbor, MI

Aug. 2023 – May 2027

EXPERIENCE

Helping Hand AI

DevOps Intern

May 2026 – Present

- Built the live camera pipeline by orchestrating **MediaMTX** to pull **RTSP** from Reolink cameras and re-serve low-latency **HLS** (1s segments) and **WebRTC/WHEP** streams to the caregiver app, with a **WHIP** path configured for two-way audio
- Designed a **FastAPI + SQLite** backend secured with **JWT (HS256)** authentication, **bcrypt** password hashing, and role-based access control, exposing a filterable, paginated alert API with an accept/dismiss/false-alarm acknowledgment workflow
- Implemented real-time tiered alert delivery over a **JWT-authenticated WebSocket** broadcast channel, persisting event type, severity, and CV confidence scores consumed by the mobile and ML/CV teams

Widget Factory (Idler YC S25)

Problem Design Engineer

March 2026 – Present

- Reviewed code changes across diverse repositories to evaluate **problem clarity, correctness, and fairness**
- Assessed **test quality** to ensure alignment between intended solutions and validation cases
- Identified **ambiguities, inconsistencies, and missing context** in problem statements and artifacts

GrimmWyrd Studios

Junior Programmer

Oct. 2024 – Feb. 2025

- Diagnosed and fixed **C++/Python** bugs and resolved deployment and hosting issues across a shared codebase in a collaborative development environment
- Implemented gameplay prototypes in **Unreal Engine 5** and created supporting 3D assets in **Blender**

PROJECTS

Network File Server | *C++, Threads, Synchronization, File Systems*

Apr. 2026

- Built a concurrent network file server supporting authenticated client requests, hierarchical file operations, and persistent file-system state
- Implemented synchronization for multi-client access, ensuring correctness under concurrent reads, writes, creates, deletes, and session handling

RISC-V Authenticated Pointers | *C++, RISC-V, Computer Architecture, Security*

Jan. 2026 – Apr. 2026

- Hardened RISC-V control/data pointers against tampering by designing a pointer-authentication extension with **PAC** generation and verification
- Evaluated security and hardware cost tradeoffs across **XOR, QARMA, CRAFT, and Ascon**-based PAC designs and documented the decode-stage integration pipeline

Out-of-Order RISC-V Processor | *Verilog, SystemVerilog, Verdi, JasperGold*

Oct. 2025 – Dec. 2025

- Implemented a high-performance **R10K-style out-of-order microprocessor** with parameterized superscalar width, reservation stations, and load/store queue
- Designed non-blocking instruction/data caches, branch prediction, early tag broadcast, return address stack, and verified correctness with custom testbenches and **JasperGold**

Embedded Systems Soccer Robots | *C, ARM Assembly, STM32L4+, CubeIDE*

Oct. 2025 – Dec. 2025

- Implemented firmware for a robot soccer game using **PS2 controllers, Zumo Robots**, LCD displays, break-beam sensors, and XBee wireless communication
- Configured STM32 peripherals and DAC-based audio output to support responsive controls, goal detection, and sound effects

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, R, Verilog, SystemVerilog, ARM Assembly

Systems/AI: RISC-V, Computer Architecture, Embedded Systems, JAX, Pallas, XLA, OpenCV

Tools & Infrastructure: FastAPI, REST APIs, WebSockets, JWT Auth, Docker / Compose, MediaMTX, RTSP/ONVIF, HLS, WebRTC, SQLite, Git, VS Code, STM32, ARM Cortex-M4, Unreal Engine 5, Blender, Figma